

New Wheels Project

Data Querying and Analytics with SQL

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Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

A series of strategic questions raised at the leadership level were identified as requiring data-driven answers. The database dump file, containing the core transactional and customer tables, was imported and queried using SQL across all key dimensions: customer distribution, vehicle preferences, revenue trends, satisfaction ratings, shipping performance, and discount patterns.

The findings have been compiled into this quarterly business report for the CEO, with raw query outputs translated into actionable observations and strategic recommendations to address the sustained decline in sales, customer satisfaction, and order volumes observed throughout the financial year.

Business Questions

Question 1:

Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
    c.state,
    COUNT(c.customer_id) AS num_customers
FROM customer_t c
GROUP BY c.state
ORDER BY num_customers DESC;
```

Output:

```
1  SELECT
2      c.state,
3      COUNT(c.customer_id) AS num_customers
4  FROM customer_t c
5  GROUP BY c.state
6  ORDER BY num_customers DESC;
```

Result: Passed

✓ Query 1

Query:

```
SELECT
    c.state,
    COUNT(c.customer_id) AS num_customers
FROM customer_t c
GROUP BY c.state
ORDER BY num_customers DESC
```

Output:

Showing first 10 rows out of 49 rows

state	num_customers
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33
Alabama	29
Washington	28

Observations and Insights:

- Texas and California jointly lead with 97 customers each, together accounting for ~20 % of the total customer base, reflecting the large population and vehicle-friendly culture in these states.
- Florida ranks third with 86 customers, as New Wheels' dominant market.
- The top 5 states account for approximately 38% of all customers, indicating a geographically concentrated customer base.
- Several states such as Vermont, Montana, and Wyoming have very few customers (less than 5 customers), suggesting untapped market potential or distribution gaps in less populated regions.

Question 2:

Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
    p.vehicle_maker,
    COUNT(DISTINCT o.customer_id) AS num_customers
FROM order_t o
JOIN product_t p ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY num_customers DESC
LIMIT 5;
```

Output:

Result: **Passed**

✓ Query 1

Query:

```
SELECT
    p.vehicle_maker,
    COUNT(DISTINCT o.customer_id) AS num_customers
FROM order_t o
JOIN product_t p ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY num_customers DESC
LIMIT 5
```

Showing 5 rows

vehicle_maker	num_customers
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- Chevrolet dominates with 83 customers, 32% more than second-placed Ford underscoring its broad appeal and strong after-market demand in the used vehicle segment.
- The top 5 makers are exclusively American domestic brands (Chevrolet, Ford, Pontiac, Dodge) with Toyota being the lone international entrant, reflecting the American pre-owned market preference.
- Pontiac and Dodge are tied at 50 customers each, competing closely for the 4th position.
- New Wheels should prioritise stocking Chevrolet and Ford inventory, as these are clearly the most in-demand makes among its customer base.

Question 3:

Which is the most preferred vehicle maker in each state?

Solution Query:

```
WITH ranked AS (
  SELECT
    c.state,
    p.vehicle_maker,
    COUNT(DISTINCT o.customer_id) AS num_customers,
    RANK() OVER (
      PARTITION BY c.state
      ORDER BY COUNT(DISTINCT o.customer_id) DESC
    ) AS rnk
  FROM order_t o
  JOIN customer_t c ON o.customer_id = c.customer_id
  JOIN product_t p ON o.product_id = p.product_id
  GROUP BY c.state, p.vehicle_maker
)
SELECT state, vehicle_maker, num_customers
FROM ranked
WHERE rnk = 1
ORDER BY state;
```

Output:

Result: Passed

✓ Query 1

Query:

```
WITH ranked AS (  
  SELECT  
    c.state,  
    p.vehicle_maker,  
    COUNT(DISTINCT o.customer_id) AS num_customers,  
    RANK() OVER (  
      PARTITION BY c.state  
      ORDER BY COUNT(DISTINCT o.customer_id) DESC  
    ) AS rnk  
  FROM order_t o  
  JOIN customer_t c ON o.customer_id = c.customer_id  
  JOIN product_t p ON o.product_id = p.product_id  
  GROUP BY c.state, p.vehicle_maker  
)  
SELECT state, vehicle_maker, num_customers  
FROM ranked  
WHERE rnk = 1  
ORDER BY state
```

state	vehicle_maker	num_customers
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Cadillac	3
Arizona	Pontiac	3
Arkansas	Chevrolet	1
Arkansas	GMC	1
Arkansas	Mitsubishi	1
Arkansas	Pontiac	1
Arkansas	Suzuki	1
Arkansas	Volkswagen	1
California	Audi	6

Observations and Insights:

- Chevrolet emerges as the dominant brand across the highest-volume states including Texas, Florida, New York, Colorado, Georgia, and Alaska.
- Several states show ties between multiple vehicle makers — California has five makers tied at 6 customers each, suggesting a highly diverse preference landscape that mirrors the state's demographic variety.
- Smaller-population states often show ties or lower counts, making the preference signal noisier and less actionable for individual state-level inventory decisions.

Question 4:

Find the overall average rating given by the customers. What is the average rating in each quarter? Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
-- Overall average rating
SELECT
    AVG(
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad'      THEN 2
            WHEN 'Okay'    THEN 3
            WHEN 'Good'     THEN 4
            WHEN 'Very Good' THEN 5
        END
    ) AS overall_avg_rating
FROM order_t;

-- Average rating by quarter
SELECT
    quarter_number,
    ROUND(AVG(
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad'      THEN 2
            WHEN 'Okay'    THEN 3
            WHEN 'Good'     THEN 4
            WHEN 'Very Good' THEN 5
        END
    ), 2) AS avg_rating
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```


Output:

Result: **Passed**

✓ Query 1

Query:

```
SELECT
  AVG(
    CASE customer_feedback
      WHEN 'Very Bad' THEN 1
      WHEN 'Bad'      THEN 2
      WHEN 'Okay'   THEN 3
      WHEN 'Good'    THEN 4
      WHEN 'Very Good' THEN 5
    END
  ) AS overall_avg_rating
FROM order_t
```

Output:

Showing 1 rows

overall_avg_rating
3.135

✓ Query 2

Query:

```
SELECT
  quarter_number,
  ROUND(AVG(
    CASE customer_feedback
      WHEN 'Very Bad' THEN 1
      WHEN 'Bad'      THEN 2
      WHEN 'Okay'   THEN 3
      WHEN 'Good'    THEN 4
      WHEN 'Very Good' THEN 5
    END
  ), 2) AS avg_rating
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_rating
1	3.55
2	3.35
3	2.96
4	2.4

Observations and Insights:

- The overall rating of 3.13 sits just above the mid-point of 3 (Okay), indicating marginal customer satisfaction that masks a deeply concerning trend.
- Q1 starts reasonably at 3.55 but deteriorates consistently each quarter, ending at an alarming 2.40 rating in Q4, a decline of over 1 full rating point across the year.
- The sharpest drop occurs between Q2 (3.35) and Q3 (2.96), crossing below 3.00 for the first time, a psychological threshold for customer sentiment.
- A continuous declining rating is a leading indicator of customer churn and negative reviews.

Question 5:

Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
    quarter_number,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS pct_very_bad,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS pct_bad,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS pct_okay,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS pct_good,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS pct_very_good
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: **Passed**

Query 1

Query:

```
SELECT
  quarter_number,
  ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) / COUNT(*), 2) AS pct_very_bad,
  ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) / COUNT(*), 2) AS pct_bad,
  ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) / COUNT(*), 2) AS pct_okay,
  ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) / COUNT(*), 2) AS pct_good,
  ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) / COUNT(*), 2) AS pct_very_good
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	pct_very_bad	pct_bad	pct_okay	pct_good	pct_very_good
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- YES, Customers are clearly getting more dissatisfied over time. In Q1, 58.71% gave Good or Very Good feedback; by Q4 this collapsed to just 20.10%.
- Negative feedback (Very Bad + Bad) surged from 22.26% in Q1 to a staggering 59.80% in Q4 — more than half of all Q4 customers were dissatisfied.
- The 'Very Bad' category shows the most dramatic shift from 10.97% in Q1 to 30.65% in Q4, nearly tripling in magnitude.
- The 'Okay' category remains relatively stable across all quarters (~19–21%), suggesting a neutral middle group that can be swung either way with targeted interventions.

Question 6:

What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(order_id) AS num_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: **Passed**

✓ Query 1

Query:

```
SELECT
    quarter_number,
    COUNT(order_id) AS num_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	num_orders
1	310
2	262
3	229
4	199

Observations and Insights:

- Orders declined every single quarter without exception, a consistent, worrying downward trend throughout the financial year.
- Total orders fell from 310 in Q1 to just 199 in Q4, a decline of 35.8% over the year.
- The steepest absolute drop was between Q1 and Q2 (-48 orders), suggesting a problem that emerged early and compounded.
- The declining order trend closely mirrors the falling customer satisfaction scores, suggesting a feedback loop, where poor service drives lower repeat purchases and fewer new customer referrals.

Question 7:

Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
WITH revenue AS (  
    SELECT  
        quarter_number,  
        ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue  
    FROM order_t  
    GROUP BY quarter_number  
)  
SELECT  
    quarter_number,  
    net_revenue,  
    LAG(net_revenue) OVER (ORDER BY quarter_number) AS prev_quarter_revenue,  
    ROUND(  
        (net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))  
        * 100.0 / LAG(net_revenue) OVER (ORDER BY quarter_number),  
    2) AS qoq_pct_change  
FROM revenue  
ORDER BY quarter_number;
```

Output:

Result: Passed

✓ Query 1

Query:

```
WITH revenue AS (  
    SELECT  
        quarter_number,  
        ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue  
    FROM order_t  
    GROUP BY quarter_number  
)  
SELECT  
    quarter_number,  
    net_revenue,  
    LAG(net_revenue) OVER (ORDER BY quarter_number) AS prev_quarter_revenue,  
    ROUND(  
        (net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))  
        * 100.0 / LAG(net_revenue) OVER (ORDER BY quarter_number),  
    2) AS qoq_pct_change  
FROM revenue  
ORDER BY quarter_number
```

Output:

Query Executed Successfully

	quarter_number	net_revenue	prev_quarter_revenue	qoq_pct_change
▶	1	18032549.90	NULL	NULL
	2	13122995.76	18032549.90	-27.23
	3	8882298.84	13122995.76	-32.32
	4	8573149.28	8882298.84	-3.48

Observations and Insights:

- Total net revenue for FY 2018 is \$48,610,993.78. The business generated 37% of its annual revenue in Q1 alone, followed by steep declines.
- The worst QoQ drop was in Q3 at -32.32%, reflecting both fewer orders and lower revenue per order.
- Q4 shows a smaller decline of -3.48% versus Q3, which may appear encouraging but is misleading — the absolute revenue level remains critically low at \$8.57M compared to Q1's \$18M.
- Net revenue formula: $\text{SUM}(\text{quantity} \times \text{vehicle_price} \times (1 - \text{discount}))$.

Question 8:

What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    quarter_number,
    ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue,
    COUNT(order_id) AS num_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: **Passed**

✓ Query 1

Query:

```
SELECT
  quarter_number,
  ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue,
  COUNT(order_id) AS num_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	num_orders
1	18032549.9	310
2	13122995.76	262
3	8882298.84	229
4	8573149.28	199

Observations and Insights:

- Both orders and revenue declined monotonically across all four quarters, confirming a systemic business decline rather than seasonal variation.
- Average revenue per order also fell from \$58,169 in Q1 to \$38,787 in Q3, suggesting customers may be purchasing lower-priced vehicles or higher discounts are being applied.
- Q4 shows a slight increase in average revenue per order (\$43,081 vs Q3's \$38,787) despite having the fewest orders, hinting that higher-value orders survived while smaller/casual purchases were lost.
- The simultaneous decline in both orders and revenue paints a complete picture of a business under stress from both volume and value perspectives.

Question 9:

What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount) * 100, 2) AS avg_discount_pct
FROM order_t o
JOIN customer_t c ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount_pct DESC;
```

Output:

Result: **Passed**

✓ Query 1

Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount) * 100, 2) AS avg_discount_pct
FROM order_t o
JOIN customer_t c ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount_pct DESC
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	avg_discount_pct
laser	64.38
mastercard	62.95
maestro	62.42
visa-electron	62.35
china-unionpay	62.22
instapayment	62.06
americanexpress	61.63

Observations and Insights:

- Laser card holders receive the highest average discount at 64.38%, followed closely by Mastercard (62.95%) and Maestro (62.42%).
- The range of discounts across card types is relatively narrow (58.4% to 64.4%) — a spread of only ~6 percentage points, suggesting discount policies are not dramatically differentiated by card type.
- Diners Club International and Solo card users receive the lowest discounts on average, possibly reflecting a smaller user base or different promotional partnerships.
- New Wheels could review discount allocation strategy if premium card types correlate with higher spending, offering them even better discounts could incentivise larger purchases and improve average order value.

Question 10:

What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
    quarter_number,
    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS avg_days_to_ship
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: Passed

✓ Query 1

Query:

```
SELECT
    quarter_number,
    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS avg_days_to_ship
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_days_to_ship
1	57.17
2	71.11
3	117.76
4	174.1

Observations and Insights:

- Shipping times deteriorated catastrophically across the year, from a manageable 57 days in Q1 to an unacceptable 174 days (nearly 6 months) in Q4.
- The sharpest single-quarter jump was in Q3 (+46.65 days), suggesting a major logistics disruption or capacity bottleneck emerged between Q2 and Q3.
- An average Q4 shipping time of 174 days is deeply problematic for a vehicle resale business, customers are waiting nearly half a year for their purchased vehicle.
- The near-perfect correlation between deteriorating shipping times and declining customer satisfaction strongly suggests that logistics failure is the primary driver of the satisfaction crisis.
- Immediate corrective action on the supply chain and shipping operations is critical to reversing the business decline.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
\$48,610,993.78	1,000	994	3.13 / 5
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
\$8,573,149.28	199	97.96 days	44.10%

Key observations from the metrics snapshot:

- Total revenue of \$48.6M was generated across 1,000 orders, averaging \$48,611 per order.
- The average customer rating of 3.13/5 indicates moderate satisfaction, with a concerning declining trend across quarters.
- Average shipping time of 97.96 days is alarmingly high, worsening significantly each quarter.
- Only 44.10% of customers gave Good or Very Good feedback, signalling a dissatisfaction problem requiring urgent attention.

Business Recommendations

Based on the comprehensive data analysis across all 10 questions, the following strategic recommendations are proposed for the leadership team at New Wheels:

1. Urgent: Fix the Shipping & Logistics Operations

- The single most critical issue is the collapse in shipping times — from 57 days in Q1 to 174 days in Q4. This directly causes the satisfaction decline and order volume drop.
- Conduct an immediate root cause analysis of the Q3 shipping spike. Review shipper capacity, warehouse operations, and routing efficiency.
- Consider renegotiating contracts with shipping partners or onboarding additional logistics providers to build redundancy.
- Set a hard KPI target to bring average shipping time below 45 days within two quarters.

2. Priority: Customer Satisfaction Recovery Plan

- With 59.8% of Q4 customers giving negative feedback, a formal satisfaction recovery plan is needed, proactive outreach to Q3/Q4 customers, service credits, or loyalty discounts.
- Implement a real-time feedback loop: send automated surveys at key touchpoints (order confirmation, shipment dispatch, delivery) and escalate negative feedback immediately.
- Target: raise average rating from 2.40 (Q4) back to at least 3.50 by end of next fiscal Q2.

3. Geographic Market Deepening in Core States

- Texas and California each have 97 customers — the highest in the country. Invest in deepening market share in these states through targeted marketing, local events, and state-specific promotions.
- Florida (86 customers) is a strong third, consider opening additional service points or shipping hubs in the South to reduce transit times for this region.
- For smaller-representation states, conduct a pilot program in 2–3 states with high vehicle ownership rates to test expansion viability.

4. Inventory Strategy: Focus on Chevrolet & Ford

- Chevrolet (83 customers) and Ford (63 customers) are the top two preferred makes by a clear margin. Ensure adequate inventory levels for both brands, especially in Texas, California, and Florida.
- Maintain a minimum stock buffer of Chevrolet vehicles to avoid stockouts given consistent demand. Consider acquiring more Chevrolet models during peak supply windows.
- For states showing diverse or tied preferences (e.g., California), maintain a broad inventory mix rather than stocking a single make.

5. Revenue Recovery Through Pricing & Discount Strategy

- Average revenue per order declined from \$58,169 in Q1 to \$38,787 in Q3. Review the discount structure. Review if discounts are being offered indiscriminately and to the right customer segments.
- Laser card holders receive the highest average discount (64.38%), evaluate whether this correlates with higher order values to justify the margin sacrifice.
- Consider tiered discount structures that reward higher-value orders rather than offering flat percentage discounts uniformly across all orders.

- With Q4 revenue at \$8.57M versus Q1's \$18M, the business needs to recover at least 50% of lost revenue over the next two quarters to remain viable.

6. Data-Driven Quarterly CEO Dashboard

- Establish a formal quarterly BI dashboard covering all 8 KPIs in the business metrics overview — revenue, orders, ratings, shipping time, and feedback distribution.
- Add leading indicators such as week-over-week order trends, shipping delays flagged in real-time, and customer feedback scores by shipper to enable proactive management.
- The CEO should receive this dashboard within the first week of each new quarter to enable timely decisions.